

**TOWARDS WRITING A BEAUTIFUL THESIS
FOR HONG KONG POLYTECHNIC UNIVERSITY DEGREE**

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CERTIFICATE OF ORIGINALITY

I hereby declare that this thesis is my own work and that, to the best of my knowledge and belief, it reproduces no material previously published or written, nor material that has been accepted for the award of any other degree or diploma, except where due acknowledgement has been made in the text.

_____ (Signed)

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Abstract

This thesis presents a template for writing a research postgraduate thesis at The Hong Kong Polytechnic University. The abstract should be a summary of the work done, consisting of 200 to 500 words.

Publications Arising from the Thesis

1. A. Author, B. Author, and C. Author. Title of the journal paper. *Name of the Journal*, 1(1):1–10, 2026.
2. B. Author and A. Author. Title of the conference paper. In *Proceedings of the Conference*, pages 1–10, 2026.

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List of Abbreviations

ATI: Activated Tag Included Method

BES: Base Event Selection Algorithm

CDF: Cumulative Distribution Function

DBEA: Delay Bounded Event Aggregation Algorithm

EAT: Event Aggregation Tree

EDT: Event Definition Tree

EEAT-C: Energy-efficient Event Aggregation Tree Building Algorithm (centralized)

EEAT-D: Energy-efficient Event Aggregation Tree Building Algorithm (distributed)

EPC C1G2: EPC Class 1 Generation 2

FPTAS: Fully Polynomial Time Approximation Scheme

GDT: Group Definition Tree

GUI: Graphical User Interface

LFBES: Latency Constraint Fixed Base Event Selection

MDS: Multidimensional Scaling

MFST: Minimum Fusion Steiner Tree

RFID: Radio Frequency Identification

RSS: Received Signal Strength

SAC: Schedule-based Anti-Collision Protocol

SACO: Schedule-based Anti-Collision Protocol (optimal)

SACWR: Schedule-based Anti-Collision Protocol (without tag replenishment)

SDP: Semidefinite Programming

SPT: Shortest Path Tree

TED: Type-based composite Event Detection

TDOA: Time Difference of Arrival

TOA: Time of Arrival

TSP: Traveling Salesman Problem

WSN: Wireless Sensor Network

Chapter 1

Introduction

This research investigates the fundamental issues for how to write a beautiful thesis.

1.1 Background

1.1.1 Beautiful Thesis

A **beautiful thesis** [1] is a Thesis written with $\text{\LaTeX}$ and San Serif font.

References

- [1] D.A. Johnson and M.M. Trivedi. Driving style recognition using a smartphone as a sensor platform. In *Intelligent Transportation Systems (ITSC), 2011 14th International IEEE Conference on*, pages 1609–1615, Oct 2011.